

Machine learning for vision

Fall 2015

Roland Memisevic

Lecture 0, January 13, 2015

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Machine learning for vision

Objectives

- ▶ Learn about the recent advances in **data driven vision**.
- ▶ Learn how to apply some **state-of-the-art learning and inference techniques in vision tasks**.
- ▶ Learn about the basics and peculiarities of **natural images statistics**.
- ▶ (+ Get some ideas about visual information processing in biological systems.)

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IFT 6268, Winter 2015

- ▶ **Classes:**
 - ▶ Tuesdays 2:30pm-4:30pm Andre-Aisenstadt 1409
 - ▶ Wednesdays 3:30pm-5:30pm Jean Coutu S1-125
- ▶ **Instructor:**

Roland Memisevic,
3349, Pav. Andre-Aisenstadt

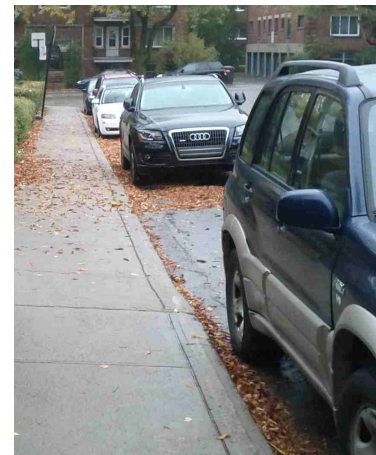
- ▶ Office hours: drop in or by appointment
- ▶ Course website:

http://www.iro.umontreal.ca/~memisevr/teaching/ift6268_2015/index.html

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Machine learning for vision

What this course is about



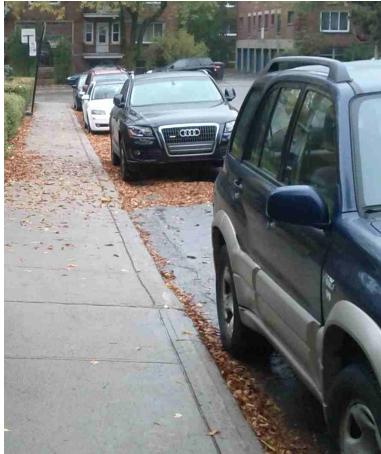
how many cars in the picture?

- ▶ Vision looks easy to humans.
- ▶ It is robust and flexible.
- ▶ It runs on fairly general-purpose hardware.

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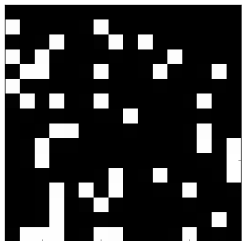
What this course is about



how many cars in the picture?

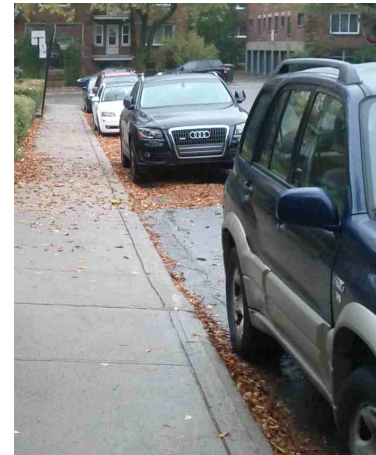
- ▶ Computer vision spent ≈ 50 years trying to mimic human vision.
- ▶ Huge inventory of tools: edge detectors, corner detectors, descriptors (eg. SIFT), optic flow, hough transform, projective geometry...
- ▶ Unfortunately, it is difficult to make these work nicely together.

A lower bound on the number of all images



- ▶ Assume your retina was only 16×16 pixels large and you could see only black and white.
- ▶ There are still $2^{16 \times 16} = 2^{256}$ *possible* images.
(=115792089237316195423570985008687907853269984665640564039457584007913129639936)
- ▶ So there are more tiny binary images than there are atoms in the universe.
- ▶ And even more large color images.

What this course is about



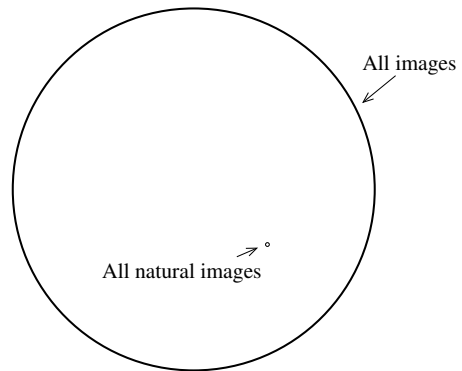
how many cars in the picture?

- ▶ Huge progress in recent years, based on a single simple idea:
- ▶ *Images are not random.*
→ Treat vision as a *statistical inference* task.

An upper bound on the number of images you will see in your life

- ▶ Assume you see 100 images per second, 3600 seconds per hour, 24 hours per day.
- ▶ This is < 10 mio images per day, or 3.65 billion images per year.
- ▶ So you will see < 300 billion images in your life and you had seen < 10 billion images when you turned 3.
- ▶ This number is tiny compared to the number of *possible* images.
- ▶ Yet, at that age you were a champion at recognizing and reasoning about unfamiliar objects.
- ▶ The number of *labeled* images is of course much smaller yet.

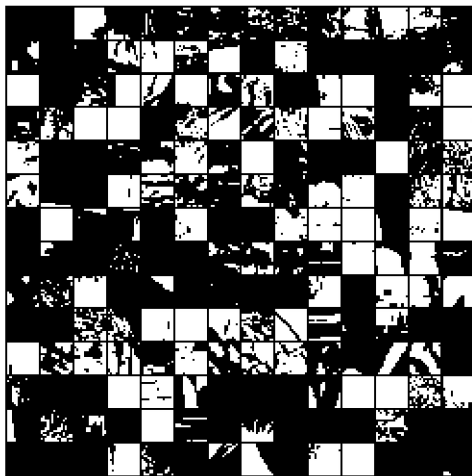
Natural images are not random



- As compared to the number of possible images, there is a diminishingly small number of *natural* images!

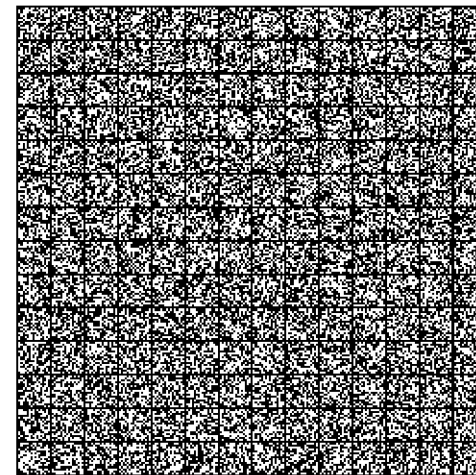
Navigation icons

Natural images (berkeley database)



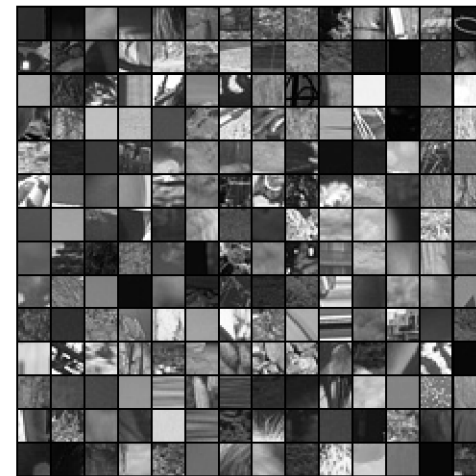
Navigation icons

Random images



Navigation icons

Natural images (grayscale)



Navigation icons

View from information theory

block images natural images



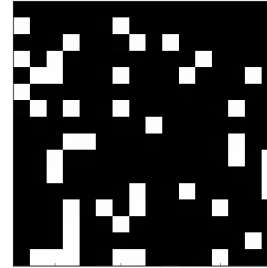
random images



Entropy

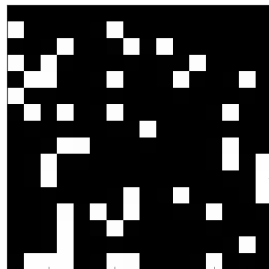
- ▶ The distribution over natural images has *low entropy*.

View from information theory



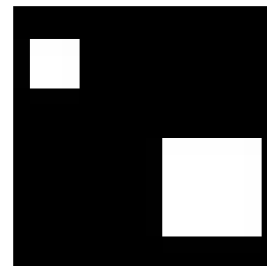
- ▶ How many bits will you need to transmit (or save) this image?

View from information theory



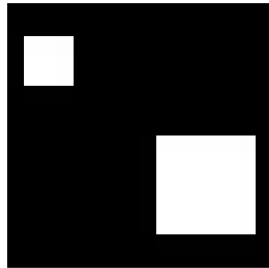
- ▶ If images are “random”, you will need 256 bits on average to transmit each.

View from information theory



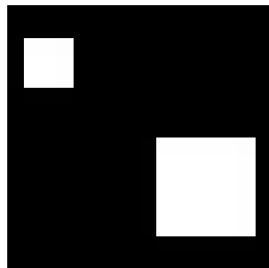
- ▶ If your images are structured, you will need much fewer bits.
- ▶ For example, what if the images contain two square blocks of random size at random locations?
- ▶ (Hyvarinen et al, 2009)

View from information theory



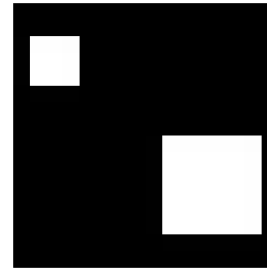
- ▶ You can transmit the upper-left corner and the bottom-right corner each with 8 bits (4 for the vertical, 4 for the horizontal direction), making it $2 \times 16 = 32$ bits for both squares.
- ▶ (It could be more efficient than that.)

View from statistics



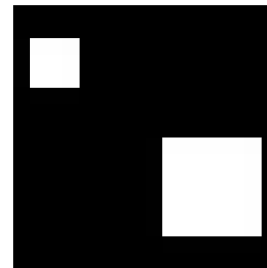
- ▶ Another way to state that the information content is small is to say that there are *dependencies* among the pixels.

View from information theory



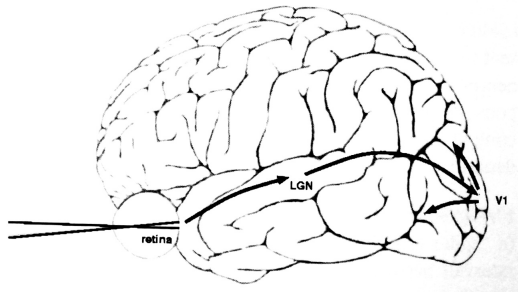
- ▶ Caveat: Neural codes, ironically, are very *high*-dimensional. It is the entropy of each individual code element that is small. This leads to *sparse* representations.

View from statistics



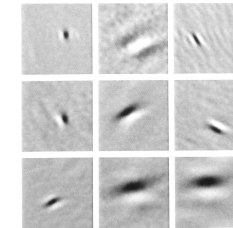
- ▶ A common way to reduce the dependencies is *Independent Components Analysis* (ICA)

View from neuroscience



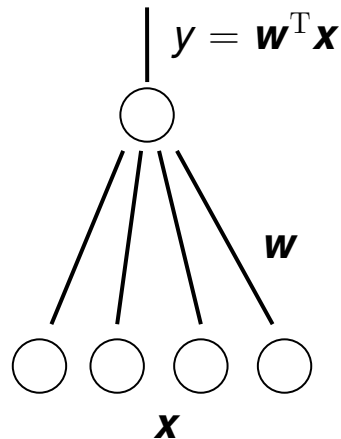
- Attneave 1954, Barlow 1961

What visual neurons like to see

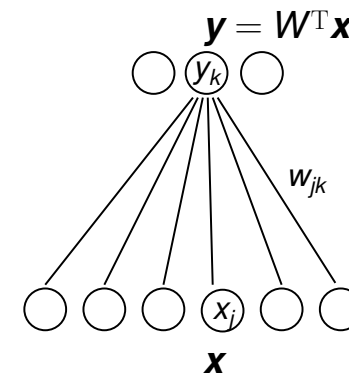


- Hubel and Wiesel, 1959

A very simple neuron abstraction



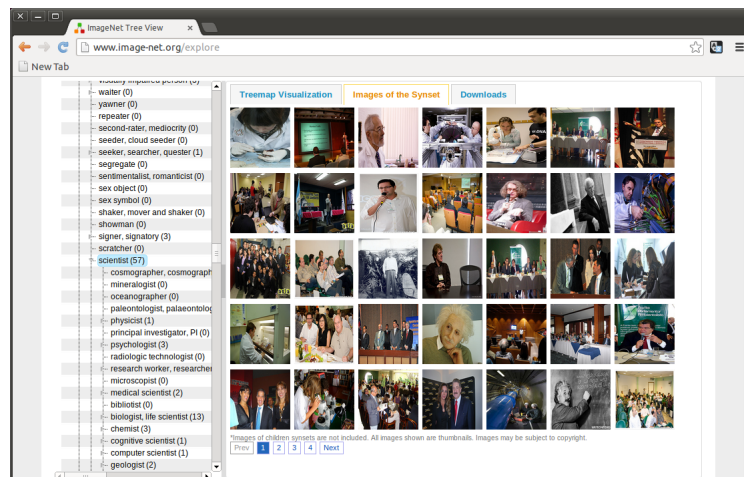
Unsupervised Learning



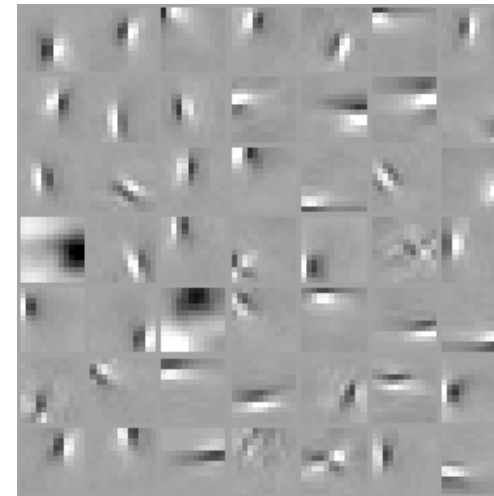
Learning criteria

- ▶ maximize independence (ica)
- ▶ minimize entropy (information theory)
- ▶ maximize sparseness (sparse coding)
- ▶ maximize probability of the data (eg. boltzmann machines, mixture models)
- ▶ learn to reconstruct from bottleneck (autoencoders, kmeans)
- ▶ supervised learning (eg. learn to classify objects)

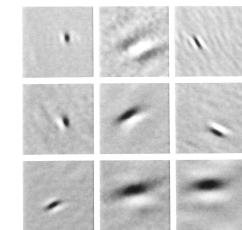
ImageNet challenge



Learned receptive fields



learned receptive fields



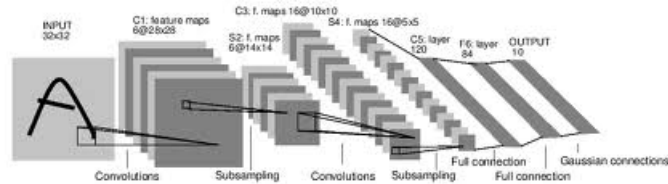
real receptive fields

ImageNet challenge

SuperVision	test-preds-141-146-2009-131-137-145-146-2011-145f	0.15315	Using extra training data from ImageNet Fall 2011 release
SuperVision	test-preds-131-137-145-135-145f.txt	0.15422	Using only supplied training data
ISI	pred_FV's_wLACs_weighted.txt	0.26172	Weighted sum of scores from each classifier with SIFT+PV, LBP+PV, GIST+PV, and CSIFT+PV, respectively.
ISI	pred_FV's_weighted.txt	0.25602	Weighted sum of scores from classifiers using each FV.
ISI	pred_FV's_summed.txt	0.26646	Naive sum of scores from classifiers using each FV.
ISI	pred_FV's_wLACs_summed.txt	0.26952	Naive sum of scores from each classifier with SIFT+PV, LBP+PV, GIST+PV, and CSIFT+PV, respectively.
			Mixed selection from High-Level SVM scores

- ▶ Krizhevsky, et al. 2012

Convolutional networks



- LeCun et al. 1998
- Fukushima 1980 (without learning)

High-level features

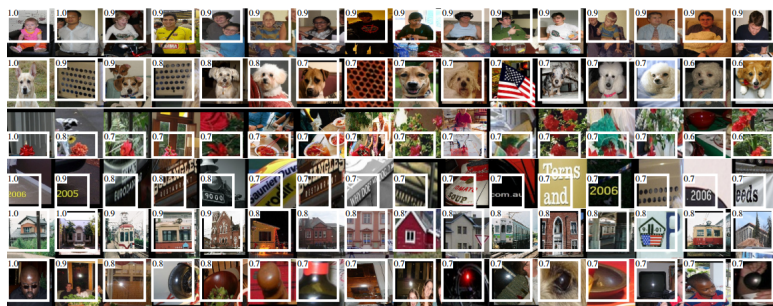


Figure 4: Top regions for six pool₇ units. Receptive fields and activation values are drawn in white. Some units are aligned to concepts, such as people (row 1) or text (4). Other units capture texture and material properties, such as dot arrays (2) and specular reflections (6).

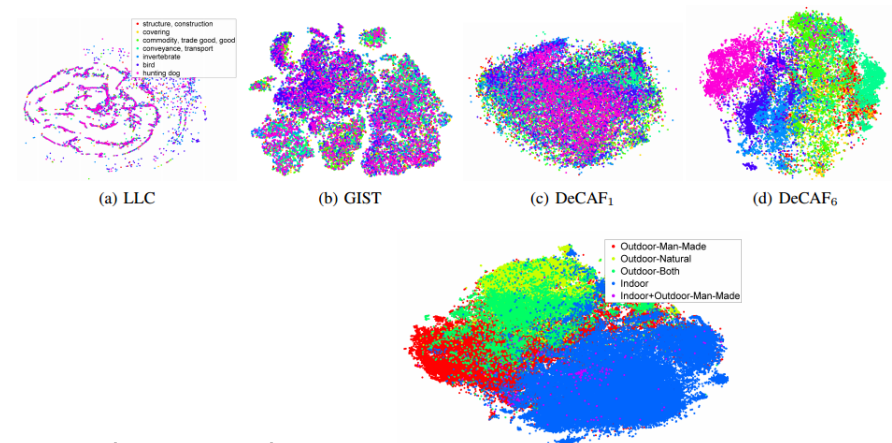
Girshick, Donahue, Darrell, Malik (!); 2014

Low-level features



- Krizhevsky, et al. 2012

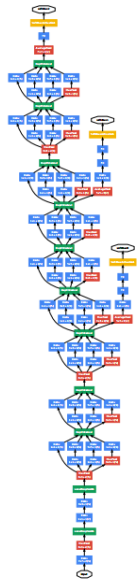
Convnet features for generic recognition



non-imagenet classes:

(Donahue et al, 2013)

GoogLeNet



- ▶ exercise in (a) scaling up, (b) unconventional neurons/architectures
- ▶ wins ImageNet 2014 with **6.66%** top-5 error rate
- ▶ vision solved?

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Vision is more than object recognition



how many cars in the picture?

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Vision is more than object recognition



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There are things images can't teach you



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There are things images can't teach you



Navigation icons: back, forward, search, etc.

There are things images can't teach you



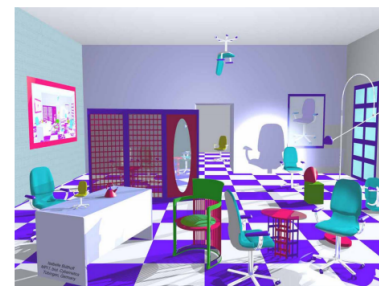
Navigation icons: back, forward, search, etc.

There are things images can't teach you



Navigation icons: back, forward, search, etc.

There are things (still) images can't teach you

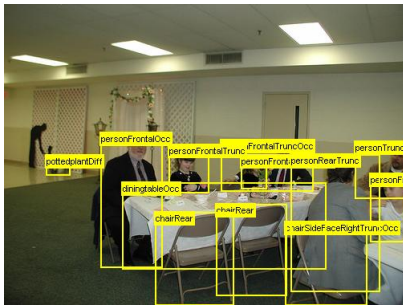


how many chairs in the picture?

(Buelthoff and Buelthoff, 2003)

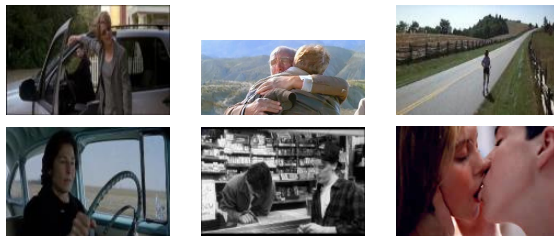
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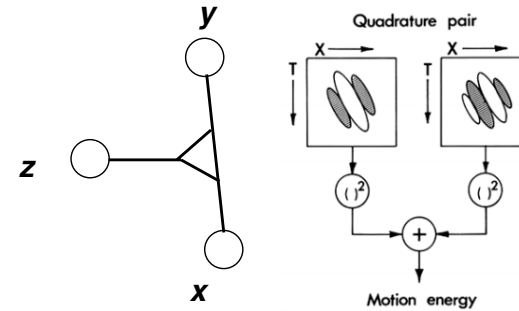
Activity recognition example



(“Hollywood 2”, Marszałek et al., 2009)

- ▶ Convolutional GBM (Taylor et al., 2010)
- ▶ hierarchical ISA (Le, et al., 2011)

A different abstract neuron



- ▶ natural way to encode time-series (video), geometry, invariance

Watching videos improves recognition

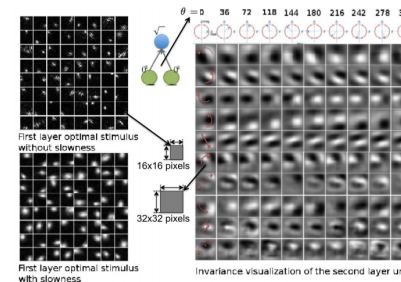


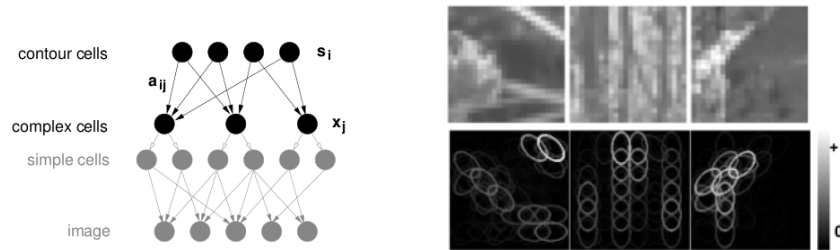
Table 3: Ave. acc. STL-10

Method	Ave. acc.
Reconstruction ICA [31]	52.9%
Sparse Filtering [40]	53.5%
SC features, K-means encoding [16]	56.0%
SC features, SC encoding [16]	59.0%
Local receptive field selection [19]	60.1%
Our result without video	56.5%
Our result using video	61.0%
Performance increase with video	+4.5%

("Deep Learning of Invariant Features via Simulated Fixations in Video", Zou, et al., 2012)

Contour coding

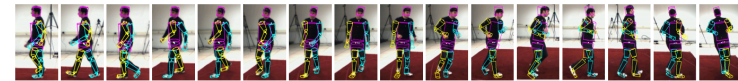
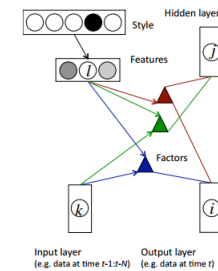
- ▶ (Hyvarinen et al., 2002)



Major conferences and journals

- ▶ **NIPS**: Neural Information Processing Systems
- ▶ **CVPR**: International Conference on Computer Vision and Pattern Recognition
- ▶ **ICCV**: International Conference on Computer Vision
- ▶ **ICML**: International Conference on Machine Learning
- ▶ **ECCV**: European Conference on Computer Vision
- ▶ **PAMI**: IEEE Transactions on Pattern Analysis and Machine Intelligence
- ▶ **Neural Computation**
- ▶ **JMLR**: Journal of Machine Learning Research

Tracking with mocap

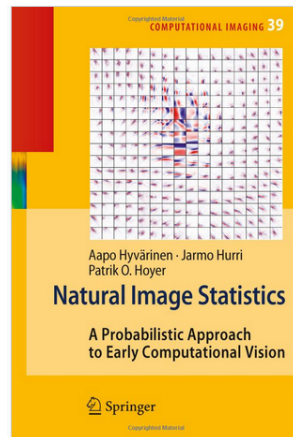


(Taylor, et al.; 2010)

Course outline

1. Fourier representations and Gabor features
2. Basic image statistics, aspects of biological vision
3. Feature learning
4. Energy models, motion, invariance
5. Advanced topics, applications, recent work

Unofficial textbook



Hyvarinen, Hurri, Hoyer: *Natural Image Statistics. A Probabilistic Approach to Early Computational Vision.*

Marking scheme

- ▶ readings (10 %)
- ▶ participation in class (20 %)
- ▶ assignments (30 %)
- ▶ term project (40 %)

Learning approach

- ▶ Readings will be posted and should be read before each class.
- ▶ Lectures will explain and motivate the concepts with real world examples.
- ▶ Student presentations of recent papers to discuss recent/novel/speculative/applied ideas.
- ▶ Several hands-on assignments to get an idea for how the methods work on actual data.
- ▶ Final projects are research based. Eg. evaluation/comparison of an approach from a recent paper, prototype/discussion of a new idea or variation of an existing one.

Relation to other courses and areas

- ▶ **Image Processing, Computer Vision:** Focus on *data* and *learning* (and *bio-inspired* as a consequence).
- ▶ **Neuroscience:** The brain (and neuroscience) is utterly complex and detailed. We will abstract away *a lot* of these details.
- ▶ **Machine Learning:** Images have *strong structure*. Black-box classifiers (like SVM) and fully Bayesian / variational methods not always the best choice.
- ▶ **Deep Learning:** Focus on images (i) to solve vision tasks, (ii) to study the models. (Eg why do DL models always learn Fourier transforms in the lowest layer?)

Tomorrow

- ▶ Review of linear algebra, stats, optimization, complex arithmetic, ...